Cofactor of 4 is $A_{31} = (-1)^{3+1} M_{31} = M_{31} = 1$

Cofactor of 1 is A_{32}

$= (-1)^{3+2} M_{32} = -M_{32} = -11$

Cofactor of 0 is A_{33}

$= (-1)^{3+3} M_{33} = M_{33} = 4$

NOTE

Value of a determinant can be obtained by using any row or column

For example

If $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$, then

$\Delta = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$ (or)

$a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$

(expanding along R_1)

$\Delta = a_{11}A_{11} + a_{21}A_{21} + a_{31}A_{31}$ (or)

$a_{11}M_{11} - a_{21}M_{21} + a_{31}M_{31}$

(expanding along C_1)

Example 1.4

Evaluate: $\begin{vmatrix} 1 & 2 & 4 \\ -1 & 3 & 0 \\ 4 & 1 & 0 \end{vmatrix}$

Solution

$\begin{vmatrix} 1 & 2 & 4 \\ -1 & 3 & 0 \\ 4 & 1 & 0 \end{vmatrix}$ = (Minor of 1) −2 (Minor of 2) + 4 (Minor of 4)

$= 1\begin{vmatrix} 3 & 0 \\ 1 & 0 \end{vmatrix} - 2\begin{vmatrix} -1 & 0 \\ 4 & 0 \end{vmatrix} + 4\begin{vmatrix} -1 & 3 \\ 4 & 1 \end{vmatrix}$

$= 0 - 0 - 52 = -52.$

1.1.4 Properties of determinants (without proof)

1. The value of a determinant is unaltered when its rows and columns are interchanged
2. If any two rows (columns) of a determinant are interchanged, then the value of the determinant changes only in sign.
3. If the determinant has two identical rows (columns), then the value of the determinant is zero.
4. If all the elements in a row (column) of a determinant are multiplied by constant k, then the value of the determinant is multiplied by k.
5. If any two rows (columns) of a determinant are proportional, then the value of the determinant is zero.
6. If each element in a row (column) of a determinant is expressed as the sum of two or more terms, then the determinant can be expressed as the sum of two or more determinants of the same order.
7. The value of the determinant is unaltered when a constant multiple of the elements of any row (column) is added to the corresponding elements of a different row (column) in a determinant.

Example 1.5

Show that

$$\begin{vmatrix} x & y & z \\ 2x + 2a & 2y + 2b & 2z + 2c \\ a & b & c \end{vmatrix} = 0$$

Solution

$$\begin{vmatrix} x & y & z \\ 2x + 2a & 2y + 2b & 2z + 2c \\ a & b & c \end{vmatrix}$$

$$= \begin{vmatrix} x & y & z \\ 2x & 2y & 2z \\ a & b & c \end{vmatrix} + \begin{vmatrix} x & y & z \\ 2a & 2b & 2c \\ a & b & c \end{vmatrix}$$

$= 0 + 0$

$= 0$

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